Project Proposal

Kindling - Open source firebase batching

What is Kindling?

Kindling is a project that is designed to solve the problem of unfriendly ways of batching data in firebase. Currently, a developer has to explicitly insert the field types and the field values directly into the firebase ui. This is slow and inconvenient for inserting large amounts of data into a database for both small scale functionality testing, and large scale performance testing. Kindling intends to solve this problem by creating a way for uploading large batches of data into firebase. Initially, this will be developed as a command line interface that will interact with the firebase api. After this prototype has been developed to prove the feasibility of the project, a visual studio code extension will be created that wraps the command line interface in a user-friendly way. This application is directly targeted at developers who use firebase regularly for both production and in development applications.

How will Kindling be developed?

Kindling has a set time scale and scope, however I do not have extensive experience in creating a visual studio code extension. This means that it best falls into the agile quadrant for development. The primary risk that occurs from this methodology is that there is a risk that a feature's difficulty and scale will be unpredictable. I have 2 plans to mitigate this issue. The first is to complete what I know first. I will implement the console application first, initially focusing on writing the input and output functionality to ensure that I have a strong base for the functionality with firebase. Then, I will implement the firebase functionality, which I have experience with but I am still not confident in it. Finally, I will create the visual code extension, which I will need to do a lot of research on. My other plan for how to mitigate these risks is to use Artificial Intelligence. I plan for AI to have a large part in my development, as this mirrors my current work environment. Artificial intelligence and effectively fill in the gaps in my knowledge, and in certain tasks exceed my knowledge greatly.

What constraints do we face?

Kindling is still in its early stages. Constraints are not fully explored and laid out. Firstly, the known constraint: Time. The deadline for the project is August 6th, however it is better to finish the application before that to give time for retrospective on the project as a whole for university. Next, an unknown constraint: Budget and Cost. The primary drivers of this constraint will be developer work hours and artificial intelligence tokens. Artificial Intelligence will be a driving development for this project, so the AI budget for the project is required to make a decision on which models will be used and when. Finally, another unknown constraint: Quality. The assumption i can make based on other projects is that quality refers to: the code being functional, works as the spec defines; the code being readable, clean indentation, naming and commenting; and ensuring the code follows object oriented principles.

How will we manage the constraints?

Firstly, Time. The deadline is set, so we have a strong, concrete framework to base our planning off. I plan to use an agile methodology for development. This will include sprint planning, sprint retrospectives, issue ticketing, and gauging effort required in story points. This fits our project well as the problem is loosely defined, there are no concrete metrics but there are developer experiences from inside our organization, and the solution is uncertain. I do not have any experience working with visual code extensions and I only have limited experience working with the firebase api. This is due to my limited experience in development (under 2 years in the workplace) so I will mitigate risks to progress by starting on the parts that I understand to ensure that there is a minimum viable product delivered by the deadline.
Next, is budget and costs. The primary costs will be developer time and Artificial Intelligence usage. Developer time shouldn't be the largest concern. Unlike previous years, development has accelerated massively due to the usage of artificial intelligence. This is one of the primary focuses I plan to write about in my analysis. However, AI is extremely expensive to use, especially if you want to use frontier models. To manage this constraint, I plan to create a plan for which models to use where. Planning and high complexity tasks will be handled by a powerful yet expensive agent, such as Claude Opus or GPT 5.5, then low complexity tasks and simple questions will be handled by a lower power, cheaper model, such as Claude Sonnet or an older GPT version. This will allow me to balance the costs of this tool while allowing myself to take advantage of the force multiplication that AI provides.
Finally, Quality constraints. This constraint is the most interesting to me, as in the work environment in my organization we have had issues with AI code quality. Usually it can make a functional application, however it inconsistently chooses between creating a temporary shortcut solution, a solution that works while not being over the top, and a solution that is written as if it were in the aerospace industry. Another problem with AI is that due to its nature, It’s good at understanding an entire class and how it connects to other classes. This means that it does not have the same ability to recognise unreadable code like a human does. It will often create bloated comments and messy variable naming. This violates the principle of quality assurance that the code must be readable. To mitigate this, I plan to do in-depth code reviews for each pull request to ensure that the code is readable, functional and maintainable, as well as writing a markdown file that sets the standards for quality inside of the program